Consumer Complaints: Data Cleaning & Analysis

Harbor & Pine Financial Services (fictional client) | 30 October 2026

Prepared by	Abhinav Yadav, Business / Data Analyst	Business owner	Head of Customer Experience & Complaints
Data source	Consumer_Complaints.xlsx (62,516 rows, 12 columns) + data dictionary	Status	Issued for review and sign-off

1. Purpose and Background

The company receives complaints about bank accounts, credit cards, mortgages, loans, credit reporting and money transfers. The complaint log is a raw spreadsheet with missing values and unclear categories, so management cannot reliably say what causes complaints, whether they are rising, or how well they are resolved. This project will **clean the data** and **analyse it** so the business can see the main complaint drivers and act on them.

2. Business Objectives

ID	Objective	How we measure success
BO-1	Create a clean, trusted complaint dataset	All records pass validation; no unexplained row loss (62,516 in = 62,516 out)
BO-2	Find what drives complaint volume	Top products and issues identified, with trend from 2017 to 2023
BO-3	Understand how complaints are resolved	Outcome mix (explanation / monetary relief / non-monetary relief) by product
BO-4	Check response timeliness	Timely-response rate by product and year
BO-5	Give actionable recommendations	At least 5 recommendations, each backed by a finding

Stakeholders

Stakeholder	Role in the project
Head of CX & Complaints (Business Owner)	Approves scope, KPI definitions and final report
Chief Compliance Officer (Sponsor)	Cares about timely and fair complaint handling
Data Governance Lead	Approves cleaning rules and data dictionary
Analyst (Abhinav Yadav)	Cleans data, analyses, reports and recommends

3. Scope

In scope	Out of scope
Profiling and cleaning all 12 fields (May 2017 to Aug 2023) Creating new fields (year, month, intake lag, resolution group) Analysis of volume, product, issue, state, channel, outcome and timeliness Cleaning log, data dictionary, dashboard/report and recommendations	Predictive modelling or text analysis (no complaint text in the data) Changing source systems or complaint processes Analysis by company, branch or customer (no such fields exist)

Assumptions: the file is complete; Complaint ID is unique; 2017 (from May) and 2023 (to 28 Aug) are partial years and must not be compared directly with full years.

4. Data Overview

Item	Detail
Records / period	62,516 complaints, 1 May 2017 to 28 Aug 2023; 51 states (50 + DC); no duplicate rows or IDs
Main fields	Complaint ID, Submitted via, Date submitted, Date received, State, Product, Sub-product, Issue, Sub-issue, Company public response, Company response to consumer, Timely response?
Product mix	Checking or savings 39.7%, Credit/prepaid card 25.9%, Credit reporting 12.3%, Mortgage 10.6%, others 11.5% (9 products)
Top issue	'Managing an account' = 15,109 complaints (24.2%)
Outcomes	Explanation only 65.7%, monetary relief 23.5%, non-monetary relief 8.4%, in progress 2.4%
Timeliness	2,403 late responses (3.9% of 61,022 with a Yes/No value)

5. Data Quality Issues Found

ID	Field	Issue	Records	Proposed fix
DQ-1	Sub-issue	Blank; mostly because Money transfer, Mortgage and Payday loan products rarely use sub-issues	10,858 (17.4%)	Fill 'Not applicable' for those products, otherwise 'Not specified'
DQ-2	Timely response?	Blank for every 'In progress' case (all 2023)	1,494 (2.4%)	Keep blank; exclude from timeliness rate
DQ-3	Company public response	Blank; and 96.5% of the rest say 'no public response'	2,175 (3.5%)	Fill 'Not recorded'; low analytical value
DQ-4	Company response	Status 'Closed' is unclear compared with 'Closed with...'	8	Group as 'Closed - unspecified' pending business answer
DQ-5	Submitted via	'Web Referral' (90) may overlap with 'Referral'; 'Email' has only 2	92+	Merge Web Referral into Referral; Email as 'Other'
DQ-6	Date submitted / received	Received date is later than submitted in 20.6% of rows (max 275 days)	12,882	Create intake_lag_days; flag lag over 30 days (435); keep original dates
DQ-7	Sub-product	Blank, all under Checking or savings account	7	Fill 'Not specified'
DQ-8	Product / dates	Very long product names; 2017 and 2023 are partial years	All	Add short product label; add partial-year flag

6. Business Requirements

ID	Requirement	Priority
BR-1	Load the file and reconcile record counts; confirm Complaint ID is unique	Must
BR-2	Treat missing values with documented rules (Section 5); do not delete records without a reason	Must
BR-3	Standardise categories (channel, response status, product short names) and keep the original values	Must
BR-4	Create derived fields: year, quarter, month, intake lag, resolution group, relief flag	Must
BR-5	Report complaint volume trends by month and year, flagging partial years	Must
BR-6	Rank products and issues by volume (Pareto) and report outcome mix and timeliness by product	Must
BR-7	Show complaints by state and by submission channel over time	Should
BR-8	Produce a dashboard/report and an insights summary with recommendations	Must
BR-9	Keep a cleaning log and make the cleaning script re-runnable	Should

KPI definitions: Timely rate = Yes / (Yes + No). Relief rate = (monetary + non-monetary relief) / total complaints. Intake lag = Date received - Date submitted (days).

7. Deliverables and Acceptance Criteria

Deliverable	Acceptance criteria
D1 Cleaned dataset (CSV/XLSX) + cleaning log + data dictionary	Row count reconciles to 62,516; no blanks left in analysis fields except documented open cases; every fix logged
D2 Analysis notebook / script (Python or Excel)	Results can be reproduced from the original file without manual steps
D3 Dashboard/report + insights summary	Every figure ties to the cleaned data; each recommendation cites a finding

8. Risks, Timeline and Sign-off

Risk / open question	Mitigation
Partial years (2017, 2023) mistaken for real decline	Flag partial years; compare like-for-like months
Meaning of status 'Closed' and 'Web Referral' unclear	Business Owner to confirm; use documented default until then
No complaint text or amount fields limit root-cause depth	State as a limitation; request extra data in a later phase

Timeline: Week 1 profiling and BRD sign-off, Week 2 cleaning, Week 3 analysis, Week 4 dashboard and recommendations.

Business Owner: _____ Date: _____	Data Governance Lead: _____ Date: _____